Fluid Dynamics

PENG

October 30, 2026

Contents

1 The Physical Properties of Fluids **1**
1.1 Solids, Liquids, and Gases 1
1.2 The Continuum Hypothesis 1

1 The Physical Properties of Fluids

Definition 1.1 (martial derivative [1])**.** *Rate of change at a point moving with the fluid locally:*

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \boldsymbol{u} \cdot \nabla = \frac{\partial}{\partial t} + [u_x u_y u_z] \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix} = \frac{\partial}{\partial t} + u_i \frac{\partial}{\partial x_i} \tag{1.1}$$

1.1 Solids, Liquids, and Gases

Definition 1.2 (simple fluid)**.** *The fluid cannot withstand any tendency by applied forces to deform it with the volume* $\equiv C$*.*

Definition 1.3 (resistance)**.** *The resistance cannot prevent the deformation such that resisting force vanishes with the rate of deformation.*

Definition 1.4 (compressibility)**.** *gases* $\gg$ *liquids*

Definition 1.5 (pressure)**.**

1.2 The Continuum Hypothesis

References

[1] George Keith Batchelor. *An introduction to fluid dynamics*. Cambridge university press, 2000. ISBN: 1139643460.